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Automated update of geometrical digital representation

Abstract

A method for updating a digital representation of an environment includes capturing an image of a portion of the environment using a change-detection device. Further, a corresponding digital data is determined that represents the portion in the digital representation of the environment. A change in the portion is detected by comparing the image with the corresponding digital data. In response to the change being above a predetermined threshold, the method includes initiating a resource-intensive scan of the portion using a scanning device, and updating the digital representation of the environment by replacing the corresponding digital data representing the portion with the resource-intensive scan.

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Background/Summary

CROSS REFERENCE TO RELATED APPLICATIONS (1) This application claims the benefit of U.S. Provisional Application Ser. No. 63/195,765, filed Jun. 2, 2021, the entire disclosure of which is incorporated herein by reference.

BACKGROUND
(1) The subject matter disclosed herein relates to use of measuring devices, such as a 3D laser scanner time-of-flight (TOF) coordinate measurement device, and in particular to automatically updating a geometrical digital representation of an environment using such measuring devices.
(2) Typically, a geometrical digital representation of an environment, such as a factory, a warehouse, a construction site, an office building, a home, or any other type of environment is captured using one or more measurement devices. The geometrical digital representation includes

representation of various features, such as objects, walls, corners, doors, windows, pipes, wires, reference points, and any other such aspects of the environment. The geometrical digital representation includes point clouds, scans, images, annotations, and other data captured by one or more measurement devices. The measuring devices can include TOF 3D laser scanners, two-dimensional (2D) scanners, cameras (such as, photogrammetry camera, wide-angle camera, mobile device, or any other image capturing device), lidar, radars, or other types of measuring devices.
(3) A measuring device such as the TOF 3D laser scanner steers a beam of light to a non-cooperative target such as a diffusely scattering surface of an object. A distance meter in the device measures a distance to the object, and angular encoders measure the angles of rotation of two axles in the device. The measured distance and two angles enable a processor in the device to determine the 3D coordinates of the target.
(4) A TOF laser scanner is a scanner in which the distance to a target point is determined based on the speed of light in air between the scanner and a target point. Laser scanners are typically used for scanning closed or open spaces such as interior areas of buildings, industrial installations and tunnels. They may be used, for example, in industrial applications and accident reconstruction applications. A laser scanner optically scans and measures objects in a volume around the scanner through the acquisition of data points representing object surfaces within the volume. Such data points are obtained by transmitting a beam of light onto the objects and collecting the reflected or scattered light to determine the distance, two-angles (i.e., an azimuth and a zenith angle), and optionally a gray-scale value. This raw scan data is collected, stored and sent to a processor or processors to generate a 3D image representing the scanned area or object.
(5) Generating an image requires at least three values for each data point. These three values may include the distance and two angles, or may be transformed values, such as the x, y, z coordinates. In an embodiment, an image is also based on a fourth gray-scale value, which is a value related to irradiance of scattered light returning to the scanner.
(6) Most TOF scanners direct the beam of light within the measurement volume by steering the light with a beam steering mechanism. The beam steering mechanism includes a first motor that steers the beam of light about a first axis by a first angle that is measured by a first angular encoder (or any other angle transducer). The beam steering mechanism also includes a second motor that steers the beam of light about a second axis by a second angle that is measured by a second angular encoder (or any other angle transducer).
(7) Many contemporary laser scanners include a camera mounted on the laser scanner for gathering camera digital images of the environment and for presenting the camera digital images to an operator of the laser scanner. By viewing the camera images, the operator of the scanner can determine the field of view of the measured volume and adjust settings on the laser scanner to measure over a larger or smaller region of space. In addition, the camera digital images may be transmitted to a processor to add color to the scanner image. To generate a color scanner image, at least three positional coordinates (such as x, y, z) and three-color values (such as red, green, blue "RGB") are collected for each data point.
(8) Reference systems provide a link between a digital world and real world using the measuring devices. The digital world includes the geometrical digital representations of the environment. For example, the reference systems can use measuring targets within an environment that can be used for planning updates to the environment. Reference systems are used to measure an environment, e.g., a building, factory or any other environment, which can be used for planning/envisioning updates to the building or factory.
(9) Accordingly, while existing reference systems are suitable for their intended purposes, what is needed is a reference system that provides an automated update between a real-world environment and the digital representations of the environment.

BRIEF DESCRIPTION

(10) According to one or more embodiments, a system for updating a digital representation of an

environment includes a change-detection device having a camera, and one or more processors responsive to executable computer instructions to perform a method. The method includes capturing an image of a portion of the environment using the change-detection device. The method further includes determining a corresponding digital data representing the portion in the digital representation of the environment. The method further includes detecting a change in the portion by comparing the image with the corresponding digital data. The method further includes in response to the change being above a predetermined threshold, initiating a resource-intensive scan of the portion using a scanning device, and updating the digital representation of the environment by replacing the corresponding digital data representing the portion with the resource-intensive scan.

(11) A method for updating a digital representation of an environment includes capturing an image of a portion of the environment using a change-detection device. The method further includes determining a corresponding digital data representing the portion in the digital representation of the environment. The method further includes detecting a change in the portion by comparing the image with the corresponding digital data. The method further includes in response to the change being above a predetermined threshold, initiating a resource-intensive scan of the portion using a scanning device, and updating the digital representation of the environment by replacing the corresponding digital data representing the portion with the resource-intensive scan.

(12) According to one or more embodiments, the digital representation of the environment comprises 3D coordinate points on surfaces of the environment.

(13) According to one or more embodiments, determining the corresponding digital data comprises locating one or more landmarks in the environment captured within the image.

(14) According to one or more embodiments, the one or more landmarks include a marker installed in the environment. According to one or more embodiments, the marker is a quick response (QR) code.

(15) According to one or more embodiments, determining the corresponding digital data representing the portion in the digital representation of the environment includes simulating a first panoramic image of the portion from a location of the change-detection device using the digital representation of the environment, and comparing the first panoramic image with the image from the change-detection device.

(16) According to one or more embodiments, initiating the resource-intensive scan of the portion includes sending a notification that includes a position in the environment, and an orientation, wherein the resource-intensive scan of the portion is to be captured by the scanning device from said position using said orientation.

(17) According to one or more embodiments, the notification further includes identification of one or more landmarks to be captured in the resource-intensive scan to register the resource-intensive scan with the digital representation.

(18) According to one or more embodiments, the resource-intensive scan is performed by an operator in response to receiving a notification from the one or more processors.

(19) According to one or more embodiments, the resource-intensive scan is performed autonomously by a robot in response to receiving a notification from the one or more processors.

(20) According to one or more embodiments, the change-detection device is a photogrammetry camera or a camera associated with a mobile phone.

(21) In one or more embodiments, the environment is a factory or building. The digital representation is used to layout/place equipment in the factory or building.

(22) In one or more embodiments, the marker includes a predetermined label that is captured in the image. The label is captured using a survey instrument in some embodiments.

(23) In one or more embodiments, the marker includes a reflective surface that is captured in the image using. The reflective surface facilitates using photogrammetry to analyze the image and determining one or more attributes of the captured portion in some embodiments.

(24) These and other advantages and features will become more apparent from the following description taken in conjunction with the drawings.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

(1) The subject matter, which is regarded as the invention, is particularly pointed out and distinctly claimed in the claims at the conclusion of the specification. The foregoing and other features, and advantages of the invention are apparent from the following detailed description taken in conjunction with the accompanying drawings in which:

(2) FIG. **1** is a perspective view of a laser scanner in accordance with an embodiment of the invention;

(3) FIG. **2** is a side view of the laser scanner illustrating a method of measurement according to an embodiment;

(4) FIG. **3** is a schematic illustration of the optical, mechanical, and electrical components of the laser scanner according to an embodiment;

(5) FIG. **4** illustrates a schematic illustration of the laser scanner of FIG. **1** according to an embodiment;

(6) FIG. **5**A illustrates a schematic illustration of the laser scanner of FIG. **1** measuring an environment according to an embodiment;

(7) FIG. **5**B illustrates a camera used to capture an image of an environment according to an embodiment;

(8) FIG. **6** illustrates an exemplary marker of FIGS. **5**A and **5**B according to an embodiment;

(9) FIG. **7**A illustrates a flow diagram illustrating a method for updating a point cloud of an environment having markers according to an embodiment;

(10) FIG. **7**B illustrates a flow diagram illustrating a method for updating a point cloud of an environment having markers according to an embodiment;

(11) FIG. **7**C illustrates a flow diagram illustrating a method for updating a point cloud of an environment having markers according to an embodiment;

(12) FIG. **8** depicts a block diagram of a measurement system to store a digital twin of an environment according to one or more embodiments;

(13) FIG. **9** depicts a flowchart of a method for updating a digital twin of an environment according to one or more embodiments;

(14) FIG. **10** depicts a flowchart of a method for detecting a change in the surroundings compared to an existing digital twin according to one or more embodiments;

(15) FIG. **11**A depicts a change-equipment device mounted on a personnel according to one or more embodiments;

(16) FIG. **11**B depicts a change-detection equipment mounted on a transportation-robot according to one or more embodiments; and

(17) FIG. **12** depicts an example of a marker/target according to one or more embodiments.

(18) The detailed description explains embodiments of the invention, together with advantages and features, by way of example with reference to the drawings.

DETAILED DESCRIPTION

(19) The technical solutions described herein relates to measuring an environment using one or more measuring devices and automatically updating a digital geometric representation of the environment. Embodiments of the technical solutions described herein provide advantages localizing a mobile photographic device, such as a cellular phone, within the environment and based on identifying one or more features, such as semantic landmarks, markers placed within the environment and their positions in a point cloud generated by the 3D measuring device.

Embodiments of the technical solutions provide advantages in providing a layout and/or placement of equipment within the environment. Embodiments of the invention provide still further advantages in updating geometric digital representations using a mobile photographic device.

(20) Referring now to FIGS. **1-3**, a laser scanner **20** is shown for optically scanning and measuring the environment surrounding the laser scanner **20**. The laser scanner **20** has a measuring head **22** and a base **24**. The measuring head **22** is mounted on the base **24** such that the laser scanner **20** may be rotated about a vertical axis **23**. In one embodiment, the measuring head **22** includes a gimbal point **27** that is a center of rotation about the vertical axis **23** and a horizontal axis **25**. The measuring head **22** has a rotary mirror **26**, which may be rotated about the horizontal axis **25**. The rotation about the vertical axis may be about the center of the base **24**. The terms vertical axis and horizontal axis refer to the scanner in its normal upright position. It is possible to operate a 3D coordinate measurement device on its side or upside down, and so to avoid confusion, the terms azimuth axis and zenith axis may be substituted for the terms vertical axis and horizontal axis, respectively. The term pan axis or standing axis may also be used as an alternative to vertical axis.

(21) The measuring head **22** is further provided with an electromagnetic radiation emitter, such as light emitter **28**, for example, that emits an emitted light beam **30**. In one embodiment, the emitted light beam **30** is a coherent light beam such as a laser beam. The laser beam may have a wavelength range of approximately 300 to 1600 nanometers, for example 790 nanometers, 905 nanometers, 1550 nm, or less than 400 nanometers. It should be appreciated that other electromagnetic radiation beams having greater or smaller wavelengths may also be used. The emitted light beam **30** is amplitude or intensity modulated, for example, with a sinusoidal waveform or with a rectangular waveform. The emitted light beam **30** is emitted by the light emitter **28** onto a beam steering unit, such as mirror **26**, where it is deflected to the environment. A reflected light beam **32** is reflected from the environment by an object **34**. The reflected or scattered light is intercepted by the rotary mirror **26** and directed into a light receiver **36**. The directions of the emitted light beam **30** and the reflected light beam **32** result from the angular positions of the rotary mirror **26** and the measuring head **22** about the axes **25** and **23**, respectively. These angular positions in turn depend on the corresponding rotary drives or motors.

(22) Coupled to the light emitter **28** and the light receiver **36** is a controller **38**. The controller **38** determines, for a multitude of measuring points X, a corresponding number of distances d between the laser scanner **20** and the points X on object **34**. The distance to a particular point X is determined based at least in part on the speed of light in air through which electromagnetic radiation propagates from the device to the object point X. In one embodiment, the phase shift of modulation in light emitted by the laser scanner **20** and the point X is determined and evaluated to obtain a measured distance d.

(23) As used herein TOF scanners may use a variety of methods for determining the distance d. These methods may include modulating the emitted light (e.g., sinusoidally) and measuring the phase shift of the returning light (phase-based scanners), or measuring the time interval between the emitted and returning light pulses (pulse-based scanners). The speed of light in air depends on the properties of the air such as the air temperature, barometric pressure, relative humidity, and concentration of carbon dioxide. Such air properties influence the index of refraction n of the air. The speed of light in air is equal to the speed of light in vacuum c divided by the index of refraction. In other words, c.sub.air=c/n. It should be appreciated that while embodiments herein may refer to a phase-shift type TOF scanner, this is for example purposes and the claims should not be so limited. A method of measuring distance based on the time-of-flight of light (whether phase or pulse based) depends on the speed of light in air and is therefore easily distinguished from methods of measuring distance based on triangulation. Triangulation-based methods involve projecting light from a light source along a particular direction and then intercepting the light on a camera pixel along a particular direction. By knowing the distance between the camera and the projector and by matching a projected angle with a received angle, the method of triangulation

enables the distance to the object to be determined based on one known length and two known angles of a triangle. The method of triangulation, therefore, does not directly depend on the speed of light in air.
(24) In one mode of operation, the scanning of the volume around the laser scanner **20** takes place by rotating the rotary mirror **26** relatively quickly about axis **25** while rotating the measuring head **22** relatively slowly about axis **23**, thereby moving the assembly in a spiral pattern. In an exemplary embodiment, the rotary mirror rotates at a maximum speed of 5820 revolutions per minute. For such a scan, the gimbal point **27** defines the origin of the local stationary reference system. The base **24** rests in this local stationary reference system.
(25) In addition to measuring a distance d from the gimbal point **27** to an object point X, the scanner **20** may also collect gray-scale information related to the received optical power (equivalent to the term "brightness.") The gray-scale value may be determined at least in part, for example, by integration of the bandpass-filtered and amplified signal in the light receiver **36** over a measuring period attributed to the object point X.
(26) The measuring head **22** may include a display device **40** integrated into the laser scanner **20**. The display device **40** may include a graphical touch screen **41**, as shown in FIG. **1**, which allows the operator to set the parameters or initiate the operation of the laser scanner **20**. For example, the screen **41** may have a user interface that allows the operator to provide measurement instructions to the device, and the screen may also display measurement results.
(27) The laser scanner **20** includes a carrying structure **42** that provides a frame for the measuring head **22** and a platform for attaching the components of the laser scanner **20**. In one embodiment, the carrying structure **42** is made from a metal such as aluminum. The carrying structure **42** includes a traverse member **44** having a pair of walls **46**, **48** on opposing ends. The walls **46**, **48** are parallel to each other and extend in a direction opposite the base **24**. Shells **50**, **52** are coupled to the walls **46**, **48** and cover the components of the laser scanner **20**. In the exemplary embodiment, the shells **50**, **52** are made from a plastic material, such as polycarbonate or polyethylene for example. The shells **50**, **52** cooperate with the walls **46**, **48** to form a housing for the laser scanner **20**.
(28) On an end of the shells **50**, **52** opposite the walls **46**, **48** a pair of yokes **54**, **56** are arranged to partially cover the respective shells **50**, **52**. In the exemplary embodiment, the yokes **54**, **56** are made from a suitably durable material, such as aluminum for example, that assists in protecting the shells **50**, **52** during transport and operation. The yokes **54**, **56** each includes a first arm portion **58** that is coupled, such as with a fastener for example, to the traverse **44** adjacent the base **24**. The arm portion **58** for each yoke **54**, **56** extends from the traverse **44** obliquely to an outer corner of the respective shell **50**, **52**. From the outer corner of the shell, the yokes **54**, **56** extend along the side edge of the shell to an opposite outer corner of the shell. Each yoke **54**, **56** further includes a second arm portion that extends obliquely to the walls **46**, **48**. It should be appreciated that the yokes **54**, **56** may be coupled to the traverse **42**, the walls **46**, **48** and the shells **50**, **54** at multiple locations.
(29) The pair of yokes **54**, **56** cooperate to circumscribe a convex space within which the two shells **50**, **52** are arranged. In the exemplary embodiment, the yokes **54**, **56** cooperate to cover all of the outer edges of the shells **50**, **54**, while the top and bottom arm portions project over at least a portion of the top and bottom edges of the shells **50**, **52**. This provides advantages in protecting the shells **50**, **52** and the measuring head **22** from damage during transportation and operation. In other embodiments, the yokes **54**, **56** may include additional features, such as handles to facilitate the carrying of the laser scanner **20** or attachment points for accessories for example.
(30) On top of the traverse **44**, a prism **60** is provided. The prism extends parallel to the walls **46**, **48**. In the exemplary embodiment, the prism **60** is integrally formed as part of the carrying structure **42**. In other embodiments, the prism **60** is a separate component that is coupled to the traverse **44**. When the mirror **26** rotates, during each rotation the mirror **26** directs the emitted light beam **30**

onto the traverse **44** and the prism **60**. Due to non-linearities in the electronic components, for example in the light receiver **36**, the measured distances d may depend on signal strength, which may be measured in optical power entering the scanner or optical power entering optical detectors within the light receiver **36**, for example. In an embodiment, a distance correction is stored in the scanner as a function (possibly a nonlinear function) of distance to a measured point and optical power (generally unscaled quantity of light power sometimes referred to as "brightness") returned from the measured point and sent to an optical detector in the light receiver **36**. Since the prism **60** is at a known distance from the gimbal point **27**, the measured optical power level of light reflected by the prism **60** may be used to correct distance measurements for other measured points, thereby allowing for compensation to correct for the effects of environmental variables such as temperature. In the exemplary embodiment, the resulting correction of distance is performed by the controller **38**.

(31) In an embodiment, the base **24** is coupled to a swivel assembly (not shown) such as that described in commonly owned U.S. Pat. No. 8,705,012 ('012), which is incorporated by reference herein. The swivel assembly is housed within the carrying structure **42** and includes a motor **138** that is configured to rotate the measuring head **22** about the axis **23**. In an embodiment, the angular/rotational position of the measuring head **22** about the axis **23** is measured by angular encoder **134**.

(32) An auxiliary image acquisition device **66** may be a device that captures and measures a parameter associated with the scanned area or the scanned object and provides a signal representing the measured quantities over an image acquisition area. The auxiliary image acquisition device **66** may be, but is not limited to, a pyrometer, a thermal imager, an ionizing radiation detector, or a millimeter-wave detector. In an embodiment, the auxiliary image acquisition device **66** is a color camera.

(33) In an embodiment, a central color camera (first image acquisition device) **112** is located internally to the scanner and may have the same optical axis as the 3D scanner device. In this embodiment, the first image acquisition device **112** is integrated into the measuring head **22** and arranged to acquire images along the same optical pathway as emitted light beam **30** and reflected light beam **32**. In this embodiment, the light from the light emitter **28** reflects off a fixed mirror **116** and travels to dichroic beam-splitter **118** that reflects the light **117** from the light emitter **28** onto the rotary mirror **26**. In an embodiment, the mirror **26** is rotated by a motor **136** and the angular/rotational position of the mirror is measured by angular encoder **134**. The dichroic beam-splitter **118** allows light to pass through at wavelengths different than the wavelength of light **117**. For example, the light emitter **28** may be a near infrared laser light (for example, light at wavelengths of 780 nm or 1150 nm), with the dichroic beam-splitter **118** configured to reflect the infrared laser light while allowing visible light (e.g., wavelengths of 400 to 700 nm) to transmit through. In other embodiments, the determination of whether the light passes through the beam-splitter **118** or is reflected depends on the polarization of the light. The digital camera **112** obtains 2D images of the scanned area to capture color data to add to the scanned image. In the case of a built-in color camera having an optical axis coincident with that of the 3D scanning device, the direction of the camera view may be easily obtained by simply adjusting the steering mechanisms of the scanner—for example, by adjusting the azimuth angle about the axis **23** and by steering the mirror **26** about the axis **25**.

(34) Referring now to FIG. **4** with continuing reference to FIGS. **1-3**, elements are shown of the laser scanner **20**. Controller **38** is a suitable electronic device capable of accepting data and instructions, executing the instructions to process the data, and presenting the results. The controller **38** includes one or more processing elements **122**. The processors may be microprocessors, field programmable gate arrays (FPGAs), digital signal processors (DSPs), and generally any device capable of performing computing functions. The one or more processors **122** have access to memory **124** for storing information.

(35) Controller **38** is capable of converting the analog voltage or current level provided by light receiver **36** into a digital signal to determine a distance from the laser scanner **20** to an object in the environment. Controller **38** uses the digital signals that act as input to various processes for controlling the laser scanner **20**. The digital signals represent one or more laser scanner **20** data including but not limited to distance to an object, images of the environment, images acquired by panoramic camera **126**, angular/rotational measurements by a first or azimuth encoder **132**, and angular/rotational measurements by a second axis or zenith encoder **134**.

(36) In general, controller **38** accepts data from encoders **132**, **134**, light receiver **36**, light source **28**, and panoramic camera **126** and is given certain instructions for the purpose of generating a 3D point cloud of a scanned environment. Controller **38** provides operating signals to the light source **28**, light receiver **36**, panoramic camera **126**, zenith motor **136** and azimuth motor **138**. The controller **38** compares the operational parameters to predetermined variances and if the predetermined variance is exceeded, generates a signal that alerts an operator to a condition. The data received by the controller **38** may be displayed on a user interface **40** coupled to controller **38**. The user interface **140** may be one or more LEDs (light-emitting diodes) **82**, an LCD (liquid-crystal diode) display, a CRT (cathode ray tube) display, a touch-screen display or the like. A keypad may also be coupled to the user interface for providing data input to controller **38**. In one embodiment, the user interface is arranged or executed on a mobile computing device that is coupled for communication, such as via a wired or wireless communications medium (e.g., Ethernet, serial, USB, Bluetooth™ or WiFi) for example, to the laser scanner **20**.

(37) The controller **38** may also be coupled to external computer networks such as a local area network (LAN) and the Internet. A LAN interconnects one or more remote computers, which are configured to communicate with controller **38** using a well-known computer communications protocol such as TCP/IP (Transmission Control Protocol/Internet({circumflex over ( )}) Protocol), RS-232, ModBus, and the like. Additional systems **20** may also be connected to LAN with the controllers **38** in each of these systems **20** being configured to send and receive data to and from remote computers and other systems **20**. The LAN may be connected to the Internet. This connection allows controller **38** to communicate with one or more remote computers connected to the Internet.

(38) The processors **122** are coupled to memory **124**. The memory **124** may include random access memory (RAM) device **140**, a non-volatile memory (NVM) device **142**, and a read-only memory (ROM) device **144**. In addition, the processors **122** may be connected to one or more input/output (I/O) controllers **146** and a communications circuit **148**. In an embodiment, the communications circuit **92** provides an interface that allows wireless or wired communication with one or more external devices or networks, such as the LAN discussed above.

(39) Controller **38** includes operation control methods embodied in application code. These methods are embodied in computer instructions written to be executed by processors **122**, typically in the form of software. The software can be encoded in any language, including, but not limited to, assembly language, VHDL (Verilog Hardware Description Language), VHSIC HDL (Very High Speed IC Hardware Description Language), Fortran (formula translation), C, C++, C#, Objective-C, Visual C++, Java, ALGOL (algorithmic language), BASIC (beginners all-purpose symbolic instruction code), visual BASIC, ActiveX, HTML (HyperText Markup Language), Python, Ruby and any combination or derivative of at least one of the foregoing.

(40) FIG. **5**A illustrates an environment **500** in which a laser scanner is utilized to make measurements according to an embodiment. A laser scanner **505**, which can be the laser scanner of FIG. **1**, can be placed at one or more designated locations within the environment **500**. The laser scanner **505** can be placed on a tripod **510** having a pan-tilt axis, or can be utilized without using the tripod **510**. The laser scanner **505** may also be placed on a movable platform, cart, or dolly. For example, the laser scanner **505** can also be installed on a factory/building ceiling and linked to a digital platform. Multiple laser scanners **505** can be used to measure environment **500**. The

measurement of the environment **500** results in the generation of an electronic model comprising a collection of 3D coordinate points on surfaces of the environment. This collection of 3D coordinate points is sometimes referred to as a point cloud.
(41) One or more targets or markers (markers) **525** (additionally illustrated in FIG. **6**) can be placed at locations with the environment **500**. The one or more markers **525** can be recognized by the laser scanner **505**. The one or more markers **525** can each have an associated Quick Response (QR) code with a unique identifier (relative to other markers placed within the environment). The location of the markers **525** may be identified with a desired level of accuracy based on the scan of the environment **500** be the laser scanner **505**. It should be appreciated that where a higher level of accuracy is desired, the locations of the markers **525** may be measured using another 3D measurement device, such as a laser tracker or a total station for example.
(42) FIG. **12** depicts an example structure of a marker **525** according to one or more aspects. The marker **525** includes among other components a portion **1201** that facilitate attaching/detaching the marker **525** from one or more other surfaces. In some aspects, the portion **1201** can be attached/detached using a screw mechanism. In other aspects, the portion **1201** can be attached/detached using magnets. The marker **525** further includes a portion **1202** that is reflective. In some cases, the portion **1202** can facilitate reflection only of certain wavelengths of light incident on the portion **1202**. The portion **1202** facilitates the marker **525** to be used for photogrammetry and other such operations. The marker **525** can further include a portion **1204** that includes a label, for example a cross that can be aimed at.
(43) The marker **525** can be used with surveying instrument (such as Total Station™) used for measurement and survey work for estimating points like even and vertical, and distance. The survey instrument, like total station, can be aimed at the portion **1204**, for example, the center point of the cross.
(44) Additionally, the marker **525** can be used for photogrammetry operations. For Photogrammetry, the marker **525** can be extracted as feature within an image that is captured if the reflective portion **1202** is captured. In some aspects, if multiple markers **525** are detected in the image, and positions of the markers can be determined, the position of the camera can be computed.
(45) In some aspects, the marker **525** can be coupled with additional objects such as a ring member that is described in the U.S. Pat. No. 10,360,727, which is incorporated herein in its entirety by reference. The ring member and the marker **525** can be coupled using the portion **1201** in some aspects, for example, by screwing the marker **525** onto the ring member (or vice versa) or coupling the marker **525** with the ring member by magnetism.
(46) Using the point cloud generated by the laser scanner **505**, a user can create a layout plan in a digital tool to mark certain points with the one or more markers **525**. The locations in which the one or more markers **525** are placed can be dependent on which points within the environment **500** the user deems as desired reference points. Technical solutions herein provide additional advantages in environments where the environment changes frequently, such as a construction site or a factory that is reconfigured often.
(47) In addition, the point cloud generated by the laser scanner **505** can include the position of one or more features of the environment **500**. For example, the features of the environment **500** can include positions of objects, corners, walls, doors, windows, furniture, beams, pipes, electrical wiring, industrial equipment, and other such features of the environment **500**.
(48) When a photogrammetry camera or a mobile device having a camera **535** (e.g., a mobile phone) captures an image of the environment **500** (see FIG. **5**B) within a field of view (FOV) **530**, the reference points can be identified. These reference points can be compared with a reference system with known measurements, i.e., a "Golden Model," or a "digital twin," a reference model or an ideal model. In an embodiment, the Golden Model is created from the point cloud generated by the laser scanner **505**. A position of the photogrammetry camera or the mobile device can be

calculated with six-degrees of freedom and stored inside the obtained image. The image can be uploaded to a digital platform, for example, a web-share cloud, and used as additional information when generating a digital twin of environment **500**, or a link between the digital twin and environment **500**. The digital twin can be a representation (e.g., a 3D model) of the factory/building having focal points (locations where markers had been placed) placed within the 3D model. In the exemplary embodiment, the digital twin is created from the point cloud generated by the laser scanner **505**, or may be the point cloud generated by the laser scanner **505**.

(49) The digital platform can display the 3D model on a computer, or to the user or another individual when located in environment **500**, i.e., on site, by projecting the locations of the one or more markers **525** within environment **500**. It should be appreciated that at least some portions of the environment **500** may change over time. For example, where the environment **500** is a manufacturing environment, the type and placement of equipment within the environment **500** may be moved or changed. Thus, embodiments provided herein provide advantages in allowing a viewer of the digital twin to view images of the environment **500** as it has existed over time, rather than a static model from when the environment **500** was scanned by the laser scanner **505**. Further, embodiments described herein facilitate to update the digital twin periodically, either on a predetermined schedule or on an ad-hoc basis. In one or more embodiments, the digital twin is updated in response to detection of one or more changes to the environment **500** since the last creation/update of the digital twin. Embodiments described herein facilitate detecting such changes using reduced or minimal resources, such as a 2D camera (instead of using an expensive and time-intensive 3D scanner). The detected change(s) is analyzed, and if the detected change(s) is at least of a predetermined magnitude, a resource-intensive scan is initiated. The resource-intensive scan is performed only on a limited portion of the environment **500**, where the change(s) is detected. The resource-intensive scan is used to update the digital twin. The specific region that is to be scanned using resource-intensive scanning is detected by determining a location and orientation of the change-detection equipment, such as the 2D camera, in the environment **500** and further identifying the corresponding region to be updated in the digital twin.

(50) FIG. **7**A is a flow diagram illustrating a method of providing image localization using a reference system. At block **705**, one or more markers can be installed at one or more locations within an environment. At block **710**, the environment can be scanned to identify and locate the one or more markers within a point cloud. Alternatively, the locations of the one or more markers can be measured, e.g., with a laser tracker, and inserted into an existing point cloud or Golden Model. At block **715**, a photogrammetry camera or a mobile device can be used to obtain one or more images of the environment. At block **720**, controller/computer can analyze the one or more images to identify each of the one or more markers in the environment and an associated location for each of the one or more markers within the point cloud or digital twin. At block **725**, controller/computer can compare the identified one or more markers to a known reference system in order to localize a position of the photogrammetry camera or the mobile device. At block **730**, the controller/computer can integrate the one or more images into a point cloud of the environment, which is then stored. At block **735**, the digital twin can be displayed and/or projected to a user. At block **740**, the digital twin can be used for planning updates to the environment, for example, a representation of equipment layout/placement, etc. It should be appreciated that this allows for the automatic or semi-automatic updating of the images within the digital twin over time as the environment changes.

(51) FIG. **7**B is a flow diagram illustrating a method of providing image localization using a reference system. At block **707**, one or more markers can be installed at one or more locations within an environment. At block **712**, the environment can be scanned to identify and locate the one or more markers within a point cloud. Alternatively, the locations of the one or more markers can be measured, e.g., with a laser tracker, and inserted into an existing point cloud or Golden Model. At block **717**, a photogrammetry camera or a mobile device can be used to obtain one or more

images of the environment. At block **722**, an image of a QR code of a marker of the one or more markers can be taken and used to assist in localizing the photogrammetry camera or the mobile device within the environment. For example, a user can scan the QR code and subsequently take pictures in the environment in which the target is in the image, or the user can take pictures in the environment and subsequently scan the QR code. At block **727**, a controller/computer can analyze the one or more images to identify each of the one or more markers in the environment and an associated location for each of the one or more markers. At block **732**, the controller/computer can compare the identified one or more markers to a known reference system in order to localize a position of the photogrammetry camera or the mobile device within the environment. At block **737**, the controller/computer can integrate the one or more images into a point cloud of the environment (digital twin), which is then stored. At block **742**, the digital twin can be displayed and/or projected to a user. At block **747**, the digital twin can be used for planning updates to the environment, for example, equipment layout/placement, etc.

(52) FIG. **7**C is a flow diagram illustrating a method of providing image localization using a reference system. At block **709**, one or more markers can be installed at one or more locations within an environment. At block **714**, the environment can be scanned to identify and locate the one or more markers within a point cloud. Alternatively, the locations of the one or more markers can be measured, e.g., with a laser tracker, and inserted into an existing point cloud or Golden Model. At block **719**, a photogrammetry camera or a mobile device can be used to obtain one or more images of the environment. At block **724**, a controller/computer can analyze the one or more images to identify each of the one or more markers in the environment and an associated location for each of the one or more markers. At block **729**, the controller/computer can compare the identified one or more markers to a known reference system in order to localize a position of the photogrammetry camera or the mobile device.

(53) At block **734**, the known reference system can be displayed and indicate possible locations for the one or more markers. At block **739**, a user can select a displayed marker, for example, by scanning a QR code of the selected marker. At block **744**, the scanner can integrate the one or more images into a point cloud of the environment (digital twin), which is then stored. At block **749**, the digital twin can be displayed and/or projected to a user. At block **754**, the digital twin can be used for planning updates to the environment, for example, equipment layout/placement, etc.

(54) Accordingly, the embodiments disclosed herein describe a system that can localize active capturing devices inside an environment with a reference system and enable users to project digital information back to reality. Localization can be established using a 2D image or 360-degree panoramic image based on known reference points. Projection of digital data back to reality can occur using, for example, a laser scanner or laser pointer.

(55) The system disclosed herein can take an image. Markers or targets of a reference system are identified as key reference points within the image. The key reference points are then compared with known coordinates of the reference system and the position of the device is calculated and stored inside the image thereby creating a digital twin.

(56) The system disclosed herein can be used in a variety of circumstances. For example, a reference system can be installed inside a factory or building. Alternatively, or in addition, the reference system can be installed remotely, for example, using cloud computing platform/technology.

(57) FIG. **8** depicts a block diagram of a measurement system to store a digital twin of an environment according to one or more embodiments. The measurement system **800** includes a computing system **810** coupled with a measurement device **820**. The coupling facilitates wired and/or wireless communication between the computing system **810** and the measurement device **820**. The measurement device **820** can include a laser tracker, a 2D scanner, a 3D scanner, or any other measurement device or a combination thereof. The captured data **825** from the measurement device **820** includes measurements of a portion from the environment. The captured data **825** is

transmitted to the computing system **810** for storage. The computing device **810** can store the captured data **825** locally, i.e., in a storage device in the computing device **810** itself, or remotely, i.e., in a storage device that is part of another computing device **850**. The computing device **850** can be a computer server, or any other type of computing device that facilitates remote storage and processing of the captured data **825**.
(58) In one or more embodiments, the captured data **825** can be used to generate a map **830** of the environment in which the measurement device **820** is being moved. The map **830** can include 3D representation, 2D representation, annotations, images, and all other digital representations of the environment **500**. The computing device **810** and/or the computing device **850** can generate the map **830**. The map **830** can be generated by combining several instances of the captured data **825**, for example, submaps. Each submap can be generated using SLAM, which includes generating one or more submaps corresponding to one or more portions of the environment. The submaps are generated using the one or more sets of measurements from the sets of sensors **822**. The submaps are further combined by the SLAM algorithm to generate the map **830**. The map **830** is the digital twin of the environment **500** in one or more embodiments.
(59) The captured data **825** can include one or more point clouds, distance of each point in the point cloud(s) from the measurement device **820**, color information at each point, radiance information at each point, and other such sensor data captured by the set of sensors **822** that is equipped on the measurement device **820**. For example, the sensors **822** can include a LIDAR **822**A, a depth camera **822**B, a camera **822**C, etc.
(60) The measurement device **820** can also include an inertial measurement unit (IMU) **826** to keep track of a pose, including a 3D orientation, of the measurement device **820**. Alternatively, or in addition, the captured data **825** the pose can be extrapolated by using the sensor data from sensors **822**, the IMU **826**, and/or from sensors besides the range finders.
(61) It should be noted that a “submap” is a representation of a portion of the environment and that the map **830** of the environment includes several such submaps “stitched” together. Stitching the maps together includes determining one or more landmarks on each submap that is captured and aligning and registering the submaps with each other to generate the map **830**. In turn, generating each submap includes combining or stitching one or more sets of captured data **825** from the measurement device **820**. Combining two or more captured data **825** requires matching, or registering one or more landmarks in the captured data **825** being combined.
(62) Here, a “landmark” is a feature that can be detected in the captured data **825**, and which can be used to register a point from a first captured data **825** with a point from a second captured data **825** being combined. For example, the landmark can facilitate registering a 3D point cloud with another 3D point cloud or to register an image with another image. Here, the registration can be done by detecting the same landmark in the two captured data **825** (images, point clouds, etc.) that are to be registered with each other. A landmark can include, but is not limited to features such as a doorknob, a door, a lamp, a fire extinguisher, or any other such identification mark that is not moved during the scanning of the environment. The landmarks can also include stairs, windows, decorative items (e.g., plant, picture-frame, etc.), furniture, or any other such structural or stationary objects. In addition to such “naturally” occurring features, i.e., features that are already present in the environment being scanned, landmarks can also include “artificial” landmarks that are added by the operator of the measurement device **820**. Such artificial landmarks can include identification marks that can be reliably captured and used by the measurement device **820**. Examples of artificial landmarks can include predetermined markers, such as labels of known dimensions and patterns, e.g., a checkerboard pattern, a target sign, spheres, or other such preconfigured markers.
(63) In the case of some of the measurement devices **820**, such as a volume scanner, the computing device **810**, **850** can implement SLAM while building the scan to prevent the measurement device **820** from losing track of where it is by virtue of its motion uncertainty because there is no presence

of an existing map of the environment (the map is being generated simultaneously). It should be noted that in the case of some types of the measurement devices **820**, SLAM is not performed. For example, in the case of a laser tracker, the captured data **825** from the measurement device **820** is stored, without performing SLAM.

(64) The measurement system **800** can be used as the reference system to create a digital twin of the environment. As can be noted, creating a geometrical digital twin of a large facility, such as a factory, a warehouse, an office building, or other such environments is a time- and cost-intensive task. However, as the environment is updated, reflecting such changes in the corresponding digital twin is cost prohibitive. A regular update of a digital twin for large facilities is not typically performed for such reasons. Hence, the digital twin is not updated in such cases resulting in an outdated digital twin of the environment **500**. Therefore, at a later stage, the digital twin cannot be used for planning updates to the environment. Instead, either the facility is scanned in entirety (again) at some point in time or scanned partially if modifications are planned, and such a partial scan is stored as a separate representation of that portion. The latter results in an unreliable digital twin, because multiple representations with different time stamps exists. In the former, between two large scanning projects of the entire facility, the real state of the facility and its digital twin diverge more and more. Multiple problems arise from this situation, such as planning based on the digital twin is unreliable, other resources such as 2D plans are used for planning, and typically, a real person has to check the scene in the real world (i.e., resources and time expensed). Eventually, in such cases, the model of the digital twin is discarded and no longer used. That is, in the end, the planners keep using their old-fashioned layout plans which have to be updated as well and does not allow 3D planning.

(65) Technical solutions described herein address such technical challenges. The technical solutions described herein facilitate detecting changes (e.g., new objects, structural changes, etc.) in the environment, and in response, triggering a new scan in the region of interest. The new scan captures the changes. The new scan is uploaded/added to the digital twin. In one or more embodiments, the new scan replaces an existing part of the digital twin. For example, the digital twin can include multiple files, each file representing a specific portion/region of the environment **500**. In such case, a specific file is replaced with the new scan. Alternatively, or in addition, the digital twin includes multiple point clouds, each point cloud representing a specific portion/region of the environment **500**. In such a case, one or more point clouds of the digital twin are replaced by the new scan. In this manner, the digital twin is partially revised and kept updated by using the technical solutions described herein.

(66) FIG. **9** depicts a flowchart of a method for updating a digital twin of an environment according to one or more embodiments. A method **900** for updating the digital twin includes recognition and locating of changes in the environment **500**, at block **902**. The locating of the change includes determining a region of interest in which the change has been detected. Further, the method **900** includes initiating a new scan of the region of interest in response, at block **904**. In one or more embodiments, the scan is performed when the detected change is “too big,” i.e., exceeds a predetermined threshold. Further, the method **900** includes updating the existing digital twin with the new scan of the region of interest, at block **906**.

(67) FIG. **10** depicts a flowchart of a method for detecting a change in the surroundings compared to an existing digital twin according to one or more embodiments. Recognizing the change is based on the existing digital twin, which is used as a ground truth to compare with the present state of the environment **500**. Method **1000** includes capturing present state of a portion of the environment **500**, at block **1082**. Getting data of the present state of the environment **500** is performed using techniques that are not resource and time intensive. For example, images taken with a change-detection equipment, for example, a low-cost consumer-grade camera can be used. In one or more embodiments, to get such images from the environment **500**, the camera can be mounted on personnel who regularly operates in the environment **500**. For example, as depicted in FIG. **11**A,

the camera **1005** can be mounted on a helmet **1002** or the clothing **1004** of the personnel **1001** in the environment. Accordingly, as the personnel **1001** moves around the environment **500**, the present state of the one or more portions in which s/he moves is captured. It is understood that the camera **1005** can be associated with the personnel **1001** in any other manner than that is depicted in FIG. **11**A. Further, it should be noted that although single personnel **1001** is depicted, in one or more embodiments, multiple personnel are equipped with the camera **1005** that capture images of the different portions of the environment **500**.
(68) In other embodiments, the camera **1005** is equipped on a mobile or transporter robot **1010** (see FIG. **11**B), for example, SPOT® from BOSTON DYNAMICS®, using a mount **1052**. It should be noted that while FIG. **11**B depicts a single camera **1005** mounted on the transporter robot **1010**, in other embodiments, the transporter robot **1010** can be fixed with multiple cameras **1005**. The cameras **1005** can be communicating with the computing device **810**, for example, to receive commands from the computing device **810** and/or to transmit captured data to the computing device **910**.
(69) The mount **1052** facilitates the camera **1005** to change pose using a motion controller (not shown). For example, the motion controller can cause the mount **1052** to rotate around its own vertical axis or horizontal axis to capture images from 360 degrees around the robot **1010**. The position of the mount **1052** in relation to the transporter robot **1010** is fixed. Accordingly, given the position of the transporter robot **1010** in the environment **500**, the position of the camera **1005** can be determined.
(70) The transporter robot **1010** includes a motion controller **1060** that controls the movement of the transporter robot **1010** through the surrounding environment. The motion controller **1060**, in one or more embodiments, sends commands to one or more motion devices **1062** of the transporter robot **1010**. The motion devices **1062** can be robotic legs, wheels, or any other such motion devices that facilitate the transporter robot **1010** to move in the surrounding environment. The transporter robot **1010** can transport the mounted camera(s) **1005** around the surrounding environment **500** accordingly. The motion device **1062** can cause the transporter robot **1010** to sway, shake, vibrate, etc., which in turn causes instability for the camera **1005** that is mounted on the transporter robot **1010**. The transporter robot **1010** can include an IMU **1064** that monitors the orientation of one or more components of the transporter robot **1010**, and particularly, the mount **1052** on which the camera **1005** is fixed. In one or more embodiments, the IMU **1064** can be queried to determine the orientation of the motion devices **1062**, the mount **1052**, or any other component of the transporter robot **1010**. In one or more embodiments, the transporter robot **1010** can be remotely controlled, for example, by the computer system **810** and/or the computer device **850**. The transporter robot **1010** can be commanded to move along a predetermined path at a predetermined frequency and speed to capture the present state of the environment **500** on a periodic basis.
(71) The camera **1005** continuously sends images to the computing system **810**, and/or the computing device **850**. In one or more examples, the camera **1005** uses a wide-angle lens to capture 360-degree images of the surrounding images.
(72) Referring to the flowchart in FIG. **10**, a location of the change-detection equipment is determined, at block **1084**. It is understood that although the embodiments described herein use a 2D camera **1005** as the change-detection equipment, in other embodiments the change-detection equipment can include a 2D scanner or any other such measurement device that cost lesser (time, resources, economic, etc.) than the scan for the digital twin (for example, 3D laser scanner). The location of the change-detection equipment is determined based on a location-detection sensor in the change-equipment device, e.g., camera **1005**. For example, the sensor can be a wireless network device that is used to connect to a network, such as a 3G/4G/5G network, a WIFI network, or any other such a network.
(73) Alternatively, the location can be determined based on the data captured by the change-equipment device **1005**. For example, images captured by the camera **1005** are analyzed to detect

one or more features, e.g., landmarks. The detected features are compared with one or more features in the images (or other type of data) in the digital twin to identify a matching region in the digital twin. For example, the images from the camera **1005** detect a refrigeration unit, an industrial robot, a fire-extinguisher, a window, or any other such landmark. Alternatively, or in addition, the detected landmarks can include artificial landmarks, such as markers, QR codes, or other such objects that are placed at predetermined locations. The corresponding portions of the digital twin are identified that include the same landmark(s). For example, if the landmark is a refrigeration unit, the portion of the digital twin that represents the room/region/portion that includes the refrigeration unit is deemed to be the location of the change-detection equipment. Alternatively, the personnel **1001**, or the transporter robot **1010** can indicate a location of the change-equipment device using an interface, such as a graphical user interface, an application programming interface, etc. Various other techniques can be used to determine a location of the change-equipment device in the environment **500**.

(74) Further, a region of interest for the captured present state is determined, at block **1086**. The region of interest of the digital twin that is identified can be a file that includes digital representation of the environment **500** that is captured from the identified location. Alternatively, or in addition, the region of interest is the digital representation of the environment **500** that includes the location. In one or more embodiments, the region of interest that is identified can be a collection of data structures from a database or any other form of stored data. For example, the collection of data structures can include one or more point clouds, one or more images, one or more annotations, or any other such data that has been captured to represent the region of interest in the digital twin. In one or more embodiments, one or more files that store such data structures are identified as the region of interest.

(75) Further, the data captured by the change-detection equipment as the present state of the region of interest is compared with the existing data of the region of interest in the digital twin, at block **1088**. For example, images from the camera **1005** are compared, using image processing algorithms, with the existing (panorama)-images from the digital twin.

(76) In one or more embodiments, performing the comparison includes generating a first panoramic image of the ground truth from the location of the change-detection equipment using the stored data in the digital twin. Generating the first panoramic image includes setting the location of the change-detection equipment as an origin and simulating a panoramic image capture from that origin in the digital twin. The first panoramic image is then compared with the panoramic image that is captured by the change-detection equipment, such as the camera **1005**. The comparison can include segmenting the two panoramic images to perform a line-detection (e.g., using an edge detection algorithm).

(77) Technical effects and benefits of the disclosed embodiments include, but are not limited to providing a system where images of an environment can be automatically or semi-automatically updated in a digital model or point cloud at a location within the digital model that matches the actual location and orientation where the image was acquired.

(78) Terms such as processor, controller, computer, DSP, FPGA are understood in this document to mean a computing device that may be located within an instrument, distributed in multiple elements throughout an instrument, or placed external to an instrument. The line segments in the two panoramic images are then compared to determine a match-score that indicates a similarity (or difference) in the two images. In one or more embodiments, an affine invariant detection can also be performed on the two panoramic images to compensate for any differences in viewpoints from which the two panoramic images are captured.

(79) In one or more embodiments, consider A to be the panoramic image of the present state of the environment **500**, i.e., the image captured by the camera **1005** mounted on the personnel **1001** or transport-robot **1010**. Further, consider that images B**1**, B**2**, . . . , Bn represent the ground truth (panorama) images in the digital twin. Changes between the two images are then detected by

comparing A pairwise with B**1**, B**2**, . . . , Bn. Affine invariant detectors can be used to match two images taken from different viewpoints. This gives a transformation from A to Bi. Note that here, A and Bi are images from the same region of interest.

(80) Using the transformation, a virtual panorama image from the digital twin (using B**1**, B**2**, . . . , Bn) from the same location as A is captured (e.g., feature available in software from FARO®). The two images from the same location are used to determine the match-score, i.e., discrepancies between the ground truth and the present state. In one or more embodiments, the match-score is determined using the Structural Similarity Index (SSIM).

(81) If the match-score is below a predetermined threshold, i.e., the difference in the two images is above a predetermined threshold, the new scan is initiated, at block **904** (FIG. **9**). Initiating a new scan includes notifying the personnel **1001** (or another personnel), the transportation-robot **1010** (or another transportation-robot) to perform a resource-intensive scan of the region of interest. In one or more embodiments, the notification includes a location to be used to perform the new scan. For example, the location of the change-detection equipment can be specified to perform the new scan. The resource-intensive scan is performed using the 3D scanner **20** or any other such measurement device(s).

(82) In one or more embodiments, the notification can also include an orientation to be used to perform the new scan. The orientation can specify a direction in which the detected change is present from the specified location. Such information can reduce the costs required to perform the new scan. Alternatively, or in addition, the notification can identify the region of interest that is to be scanned. The new scan can then be performed to ensure that the region of interest is captured in the new scan.

(83) In one or more embodiments, the notification is an electronic message, such as an email, an instant message, or any other type of electronic message to an operator that performs the resource-intensive new scan. The electronic message can include specific coordinates and the orientation from which to perform the new scan. Alternatively, or in addition, the electronic message includes one or more images of the region of interest that is to be scanned. In one or more embodiments, the notification includes one or more commands that are sent to an automated transport-robot **1010** that is equipped with the resource-intensive scanning equipment, such as the 3D scanner. The one or more commands instruct the transport-robot **1010** to be located at a particular position in the environment **500** and at a particular orientation to capture the region of interest.

(84) In one or more embodiments, the notification indicates one or more landmarks that are to be captured in the new scan. The landmarks indicated are determined based on those that are used to register the existing data in the digital twin. For example, the existing data in the digital twin for the region of interest can use a specific set of landmarks to register one or more point clouds representing the region of interest with the rest of the point clouds in the digital twin. In such a case, the landmarks are identified so that the new scan also includes the landmarks for registering the new scan with the digital twin.

(85) Further, at block **906**, the new scan of the region of interest is captured and incorporated into the digital twin. The digital twin is updated with the new scan. The updating can include replacing the existing data of the region of interest with the new scan. The new scan is registered with the rest of the digital twin using the one or more landmarks in the new scan. The landmarks include those that are identified in the notification. The registration can be performed using one or more known techniques such as, Cloud2Cloud registration, or any other registration of point clouds, images, etc.

(86) In this manner, the digital twin is updated in a continuous manner, and in parts as changes to the environment **500** are made. The digital twin can, accordingly, be readily used for planning and other purposes. The embodiments of the technical solutions described herein facilitate such a continuous update of the digital twin on a periodic basis or ad hoc when a change is detected. Because the change-detection is performed with less resources than the resource-intensive scanning process, the digital twin can be maintained in an updated state in this manner.

(87) While the invention has been described in detail in connection with only a limited number of embodiments, it should be readily understood that the invention is not limited to such disclosed embodiments. Rather, the invention can be modified to incorporate any number of variations, alterations, substitutions or equivalent arrangements not heretofore described, but which are commensurate with the spirit and scope of the invention. Additionally, while various embodiments of the invention have been described, it is to be understood that aspects of the invention may include only some of the described embodiments. Accordingly, the invention is not to be seen as limited by the foregoing description, but is only limited by the scope of the appended claims.

Claims

1. A camera system for updating a three-dimensional (3D) digital representation of an environment, the camera system comprising: a change-detection device having a camera that captures two-dimensional (2D) images of environments; a scanning device that captures 3D coordinate data of objects within environments that are used to generate 3D digital representations of objects; and at least one processor responsive to executable computer instructions to exclusively initiate: capturing a 2D image of at least a portion of an environment using the change-detection device; identifying the portion in a 3D digital representation generated from the 3D coordinate data measured by the scanning device that corresponds to the portion in the 2D image captured using the change-detection device; comparing a difference between the 2D image of the portion and the 3D digital representation of the portion against a predetermined threshold in order to detect a change in the portion of the environment between a time of the capturing and a time of the 3D digital representation; and in response to the change being above the predetermined threshold: capturing a new scan of the portion using the scanning device; and updating the 3D digital representation of the portion by replacing the portion of the 3D digital representation with the new scan of the portion; and outputting an updated 3D digital representation of the environment based on the updating.
2. The camera system of claim 1, wherein the 3D digital representation of the environment comprises 3D coordinate points on surfaces of objects within the environment.
3. The camera system of claim 1, wherein identifying the portion in the 3D digital representation further comprises identifying at least one landmark in the environment captured within the 2D image and a digital twin of the at least one landmark in the 3D digital representation.
4. The camera system of claim 3, wherein the at least one landmark includes a marker installed in the environment.
5. The camera system of claim 4, wherein the marker is a quick response (QR) code.
6. The camera system of claim 4, wherein the marker comprises a label having at least one of a known dimension and a known pattern that is captured in the 2D image.
7. The camera system of claim 4, wherein the marker comprises a reflective surface that is captured in the 2D image.
8. The camera system of claim 1, wherein identifying the portion in the 3D digital representation further comprises: simulating a panoramic image of the portion from a location of the change-detection device using the 3D digital representation of the environment; and comparing the panoramic image with the 2D image from the change-detection device.
9. The camera system of claim 1, wherein capturing the new scan of the portion comprises: sending a notification that includes a position and an orientation within the environment, wherein the new scan of the portion is captured by the scanning device from the position using the orientation.
10. The camera system of claim 9, wherein the notification further comprises identification of at least one landmark to be captured in the resource-intensive scan to register the resource-intensive scan with the 3D digital representation.
11. The camera system of claim 9, wherein the new scan is performed by the scanner in response to a manual initiation by an operator receiving the notification from the at least one processor.

12. The camera system of claim 9, wherein the new scan is performed by the scanner and is autonomously initiated by a robot in response to receiving the notification from the at least one processor.
13. The camera system of claim 1, wherein the change-detection device comprises at least one of: a photogrammetry camera; and a camera associated with a mobile phone.
14. The camera system of claim 1, wherein the environment comprises at least one of: a factory; and a building.
15. The camera system of claim 14, wherein the updated 3D digital representation is used to layout a place for new equipment in the environment.
16. A method performed exclusively by a processor of a camera system to automatically update three-dimensional (3D) digital representations of an environments, the method comprising: capturing a two-dimensional (2D) image of a portion of an environment using a change-detection device of the camera system; identifying the portion in a 3D digital representation generated from measurement data determined from a scan of the environment by a scanning device of the camera system, which corresponds to the portion in the 2D image captured using the change-detection device; comparing the 2D image of the portion of the environment and the portion in the 3D digital representation against a predetermined threshold to detect a change in the portion of the environment between a time of the 2D image and a time of the 3D digital representation; in response to the change exceeding the predetermined threshold: capturing a new scan of the portion using a scanning device; updating the 3D digital representation of the environment by replacing the portion in the 3D digital representation with the new scan; and outputting an updated 3D digital representation of the environment based on the updating.
17. The method of claim 16, wherein the 3D digital representation comprises 3D coordinate points measured on surfaces of the environment.
18. The method of claim 16, wherein identifying the portion in the 3D digital representation comprises locating at least one landmark within the 2D image.
19. The method of claim 18, wherein the at least one landmark each includes an artificial marker placed in the environment.
20. The method of claim 16, wherein comparing the 2D image of the portion further comprises: simulating a panoramic image of the portion from a location of the change-detection device using the 3D digital representation of the environment; and comparing the panoramic image with the 2D image captured by the change-detection device.
21. The method of claim 16, wherein capturing the new scan comprises: sending a notification that includes a position in the environment and an orientation, whereby the new scan of the portion is captured by the scanning device from the position using the orientation.
22. The method of claim 16, wherein the 2D camera and the scanning device share an optical axis.